MPC + INDI versus MPC + PID · butterfly comparison

3 m/s: both complete six eights. 5 m/s: neither completes; the failure modes differ.

Window / measurement	MPC + INDI	MPC + PID
3 m/s · six-eights position RMS / max	3.80 / 7.73 cm	5.35 / 11.55 cm
3 m/s · rate-target error RMS norm	0.296 rad/s	0.426 rad/s
3 m/s · completed / requested	6 / 6 · normal landing	6 / 6 · normal landing
5 m/s · first 9.98 s position RMS / max	7.61 / 17.57 cm	8.29 / 21.82 cm
5 m/s · first 9.98 s rate-error RMS norm	1.809 rad/s	0.635 rad/s
5 m/s · full attempt completed	1.554 / 6 eights	2.068 / 6 eights
5 m/s · initiating failure	Native allocator fault latch	Stale MPC result tick
5 m/s · subsequent outcome	PID fallback + emergency landing	Emergency brake then hard altitude stop

Matched: 4 x 3 m XY footprint, signed Z amplitude 0.3 m, three A/B cycles, six eights and speed-specific phase schedule.

Main 5 m/s scores use only the first 9.98 s, before native INDI fallback and alignment uncertainty. Later PID continuation is shown separately.

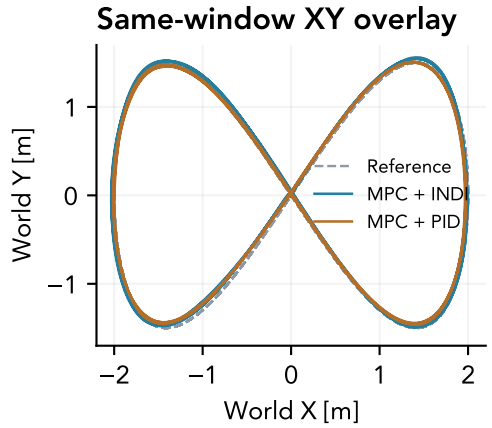
Rate error uses FC-delivered frontend target minus the same early gyro, in the same native-row clock. Internal feedback filters and actual commands differ.

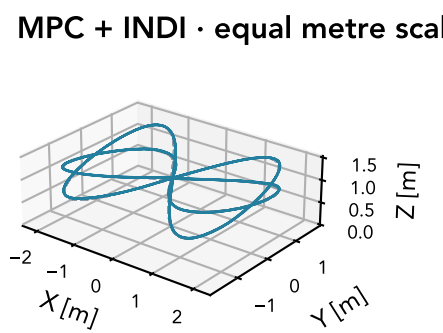
This compares paired systems: identified-rate prediction for MPC + INDI versus ideal-rate prediction for MPC + PID. Position RMS is not solely an inner-gain effect.

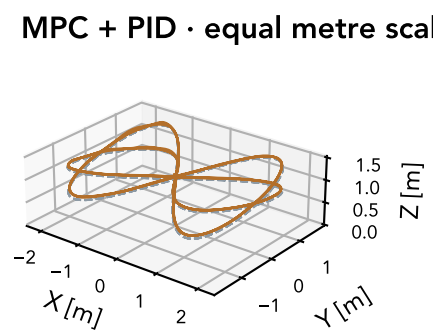
Recorded trajectories and native samples only; no added filters, resampling or fitted shifts. No MPC + PID 4 m/s flight is available for a paired claim.

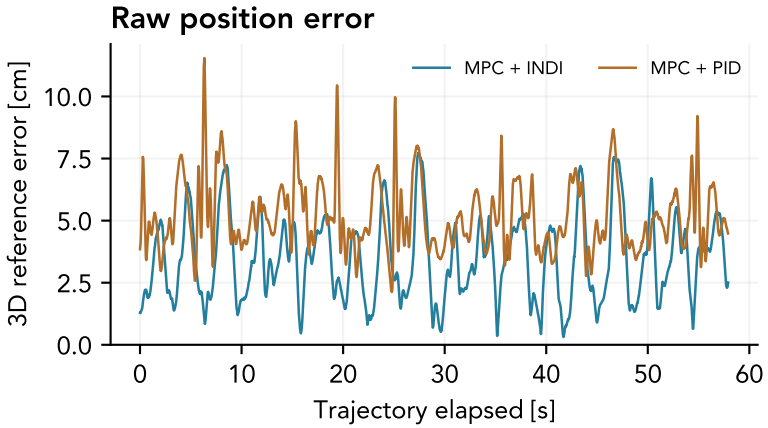
3 m/s · direct MPC + INDI / MPC + PID comparison

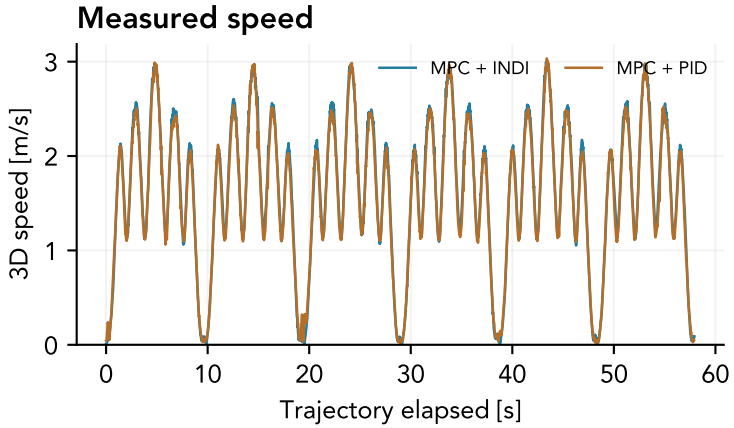
Both complete six eights; identical plot scales.

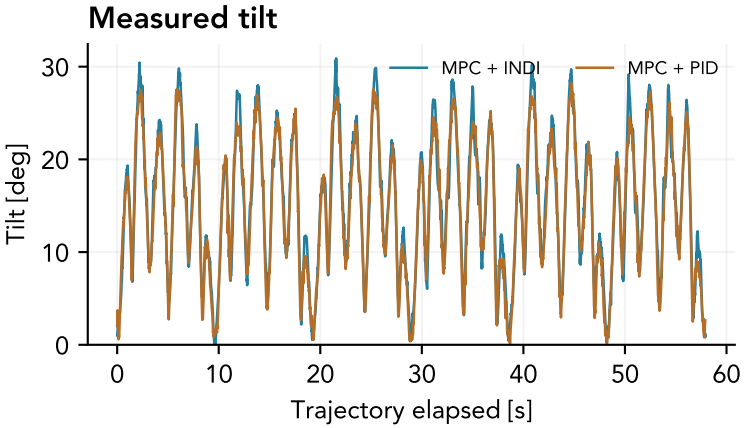












Position RMS / max [cm] · MPC + INDI 3.80 / 7.73 | MPC + PID 5.35 / 11.55

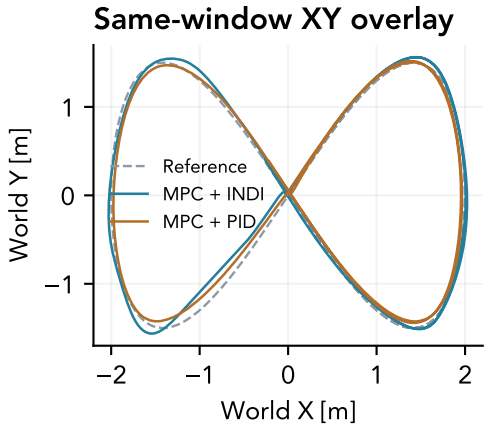
Native rate-error RMS norm [rad/s] · MPC + INDI 0.296 | MPC + PID 0.426. Different actual commands / internal feedback filters are retained.

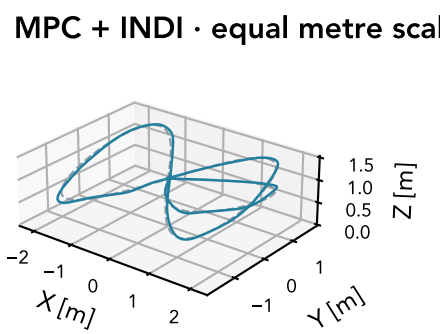
Tracking windows are approximately 57.92 s. MPC + INDI has lower position and frontend-target tracking RMS in these two recorded attempts.

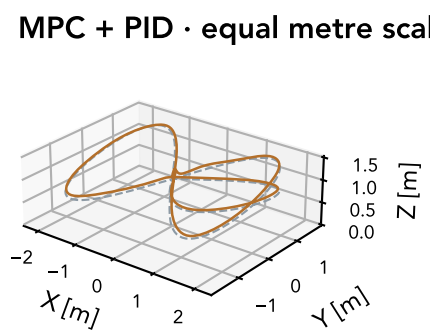
No visual filtering or fitted alignment. Errors use each recorded reference; fresh-ground datum differences are retained in world XYZ.

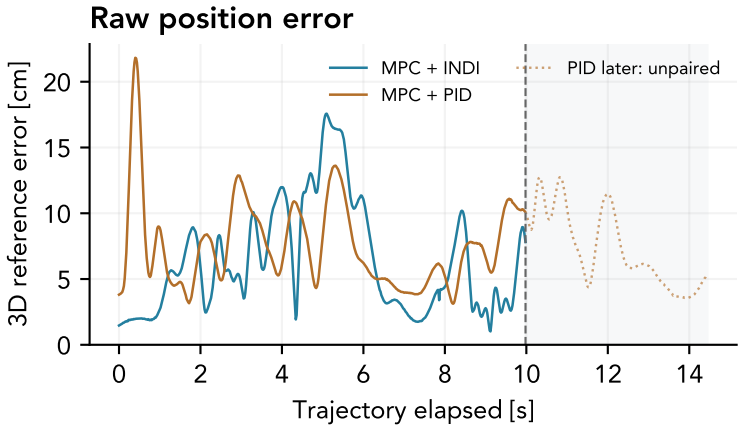
5 m/s · direct MPC + INDI / MPC + PID comparison

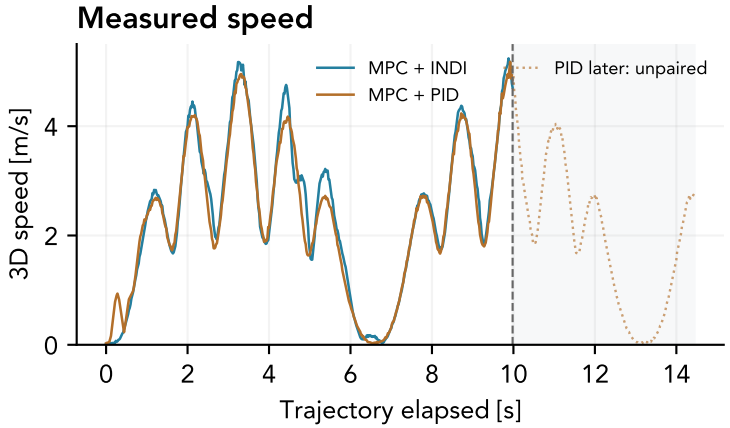
Fair scores / upper paths: first 9.98 s only. Grey region below is later MPC + PID continuation, without a paired INDI sample.

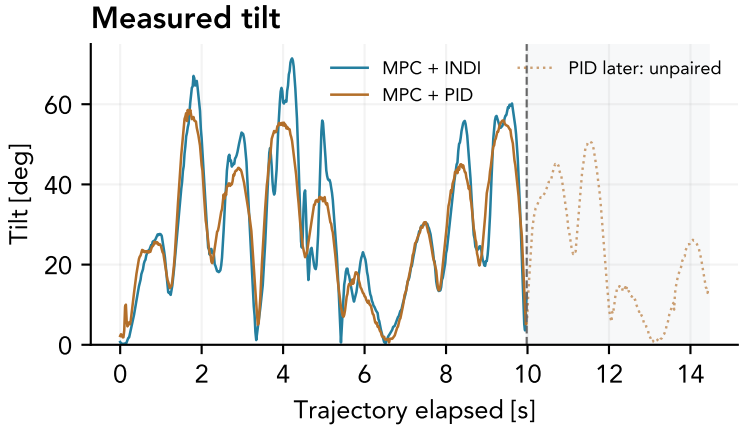












Position RMS / max [cm] · MPC + INDI 7.61 / 17.57 | MPC + PID 8.29 / 21.82

Native rate-error RMS norm [rad/s] · MPC + INDI 1.809 | MPC + PID 0.635. Different actual commands / internal feedback filters are retained.

Full tracking ends: MPC + INDI 10.14 s (1.554 eights); MPC + PID 14.44 s (2.068 eights). Neither attempt completes the requested six.

No visual filtering or fitted alignment. Errors use each recorded reference; fresh-ground datum differences are retained in world XYZ.

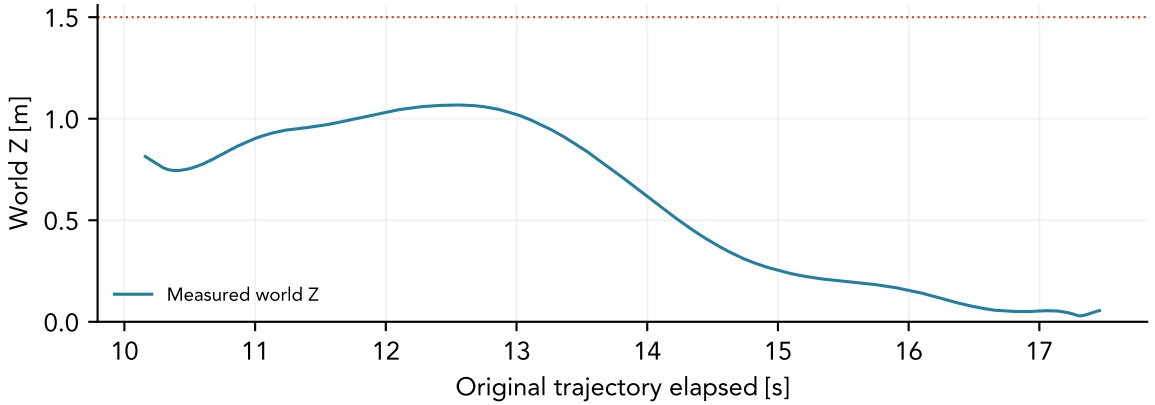
5 m/s · different initiating faults, different recovery states

Neither attempt completed. Component reliability and recovery must be assessed separately from common-window tracking RMS.

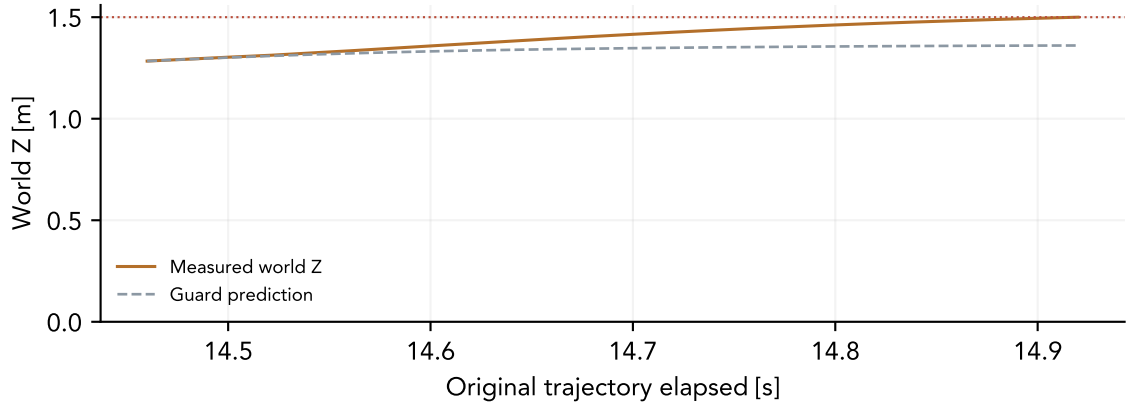
Native allocator: the FC INDI block maps requested thrust and rotational acceleration to four bounded rotor-speed targets.
A fault means no solution was accepted; hitting a bound alone can still give a valid bounded solution.

Component / chronology	MPC + INDI attempt	MPC + PID attempt
Initiating event	Native allocator fault latch	MPC result tick became stale
Host tracking exit	10.160 s · 1.554 / 6 eights	14.460 s · 2.068 / 6 eights
Evidence	Rate overshoot; bounded / motor-3 limit precursor	Valid solves; pipeline finished after 20 ms selection
Limit of evidence	Triggering 1 kHz reject not retained at 250 Hz	~0.24 ms optimizer; most delay outside optimizer
Emergency / outcome	2 rad/s; native PID fallback; landed after 7.30 s	2 rad/s; stayed PID; ceiling crossed after 0.460 s

MPC + INDI · PID fallback recovery



MPC + PID · emergency height rise



Recovery initial states differ: INDI run Z 0.814 m, vz −0.417 m/s; PID run Z 1.284 m, vz +0.461 m/s. Recovery outcomes do not isolate inner-loop quality.

Next: profile the full 20 ms result pipeline; validate PID collective / vertical response and ceiling margin; capture the initiating INDI allocator cycle.

The PID rotor-based thrust estimate is model-derived, not measured force. Native timing increments near the INDI fault do not prove causality.

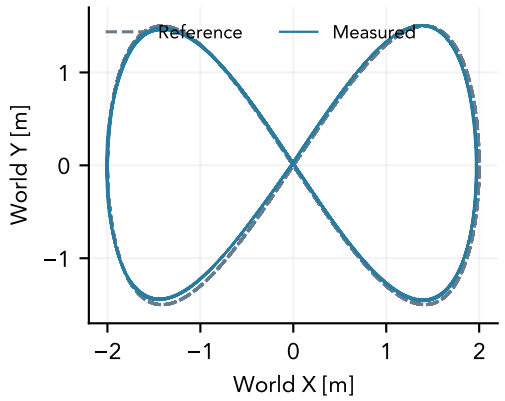
No controller, firmware or guard changed for this analysis. Detailed per-run paths, native measurements, recovery and timing follow in the appendix.

MPC + PID · 3 m/s butterfly · 6 complete eights

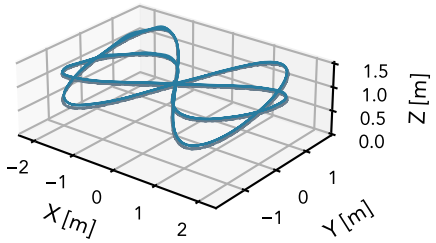
12 Nov 2026 10:21:18 UTC | 4 x 3 m XY · three A/B cycles · signed Z amplitude 0.3 m · world center Z 1.028 m

Tracking RMS 5.35 cm · max 11.55 cm | peak 3.035 m/s · 28.27° | 6.000 / 6 eights

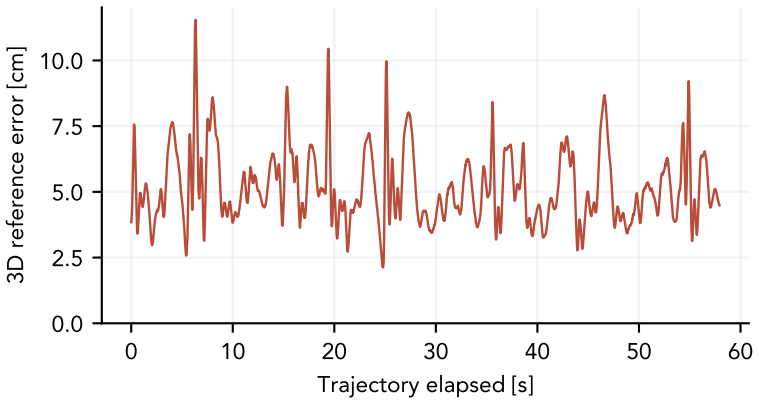
Raw XY path



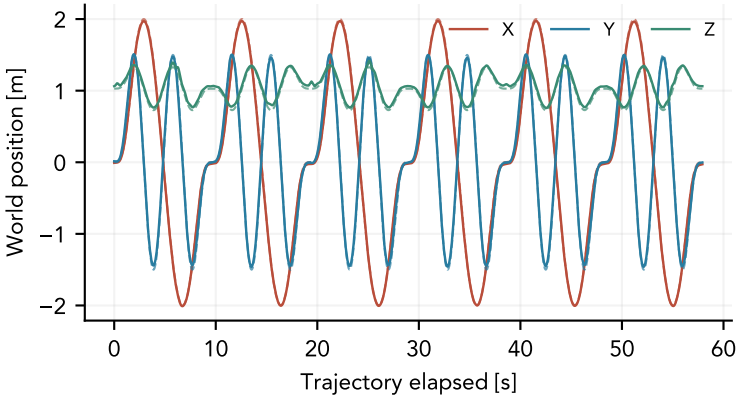
World XYZ · equal metre scale



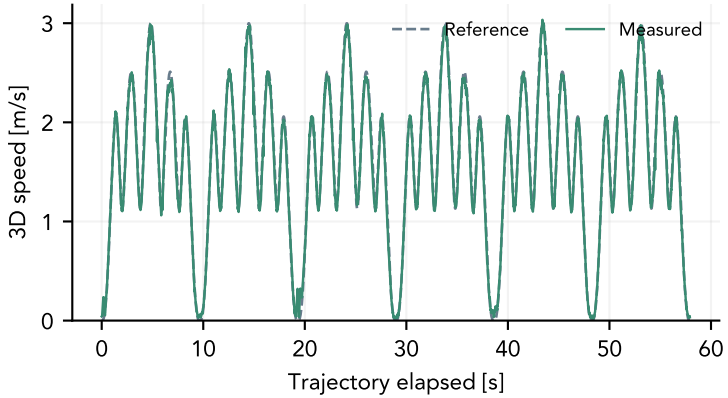
Raw error to selected reference



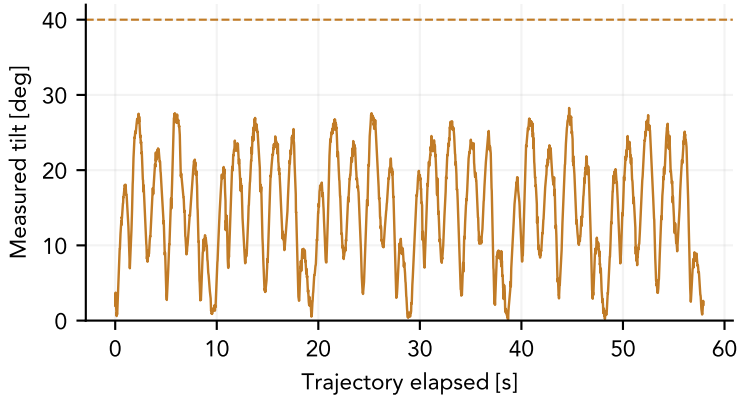
XYZ · solid measured / dashed reference



Recorded velocity state



Tilt · soft guard 40°



Per-eight RMS / max (cm): 1: 5.77 / 11.55 | 2: 5.48 / 9.00 | 3: 5.47 / 10.45 | 4: 5.08 / 8.42 | 5: 5.16 / 8.68 | 6: 5.09 / 9.21

MPC: 2897 healthy ticks · 0 RTI rescues · 0 invalid | mocap age median / max 2.96 / 3.80 ms

Native PID stayed healthy through tracking; no host envelope landing. Full six-eights trajectory followed by normal return, hold and landing.

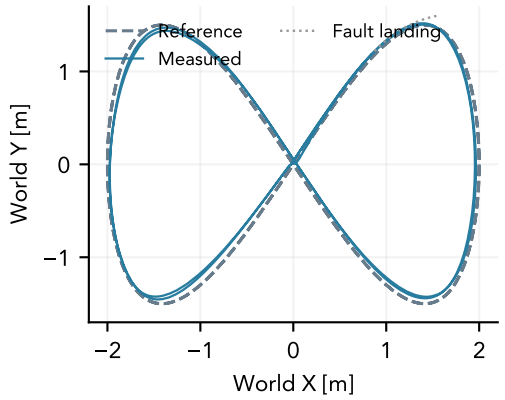
Actual recorded samples, no visualization filtering or time shift. State velocity uses the live causal estimator. FC native-rate audit follows separately.

MPC + PID · 5 m/s butterfly · partial run · hard ceiling during emergency brake

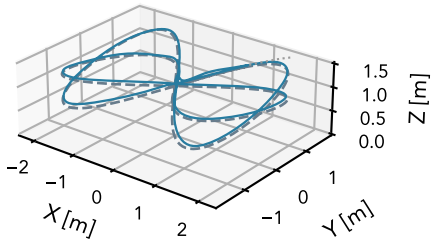
12 Nov 2026 10:23:05 UTC | 4 x 3 m XY · three A/B cycles · signed Z amplitude 0.3 m · world center Z 1.028 m

Tracking RMS 8.15 cm · max 21.82 cm | peak 5.178 m/s · 58.60° | 2.068 / 6 eights

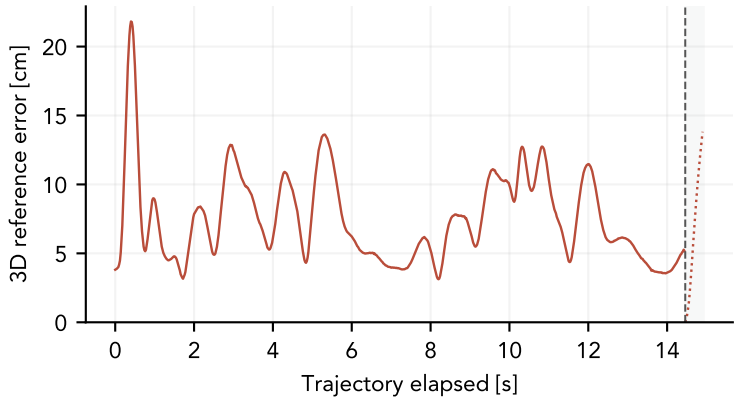
Raw XY path



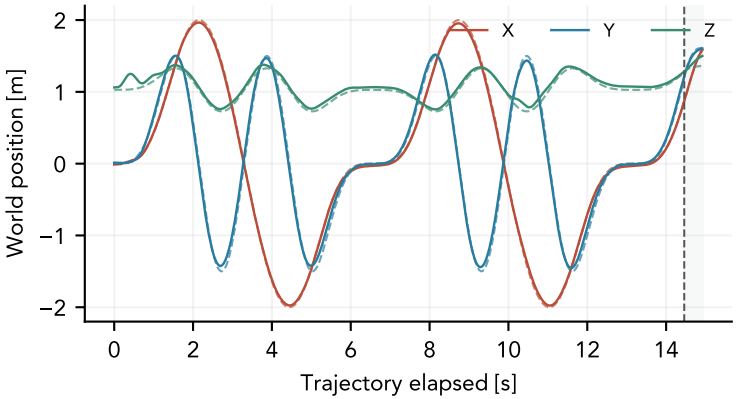
World XYZ · equal metre scale



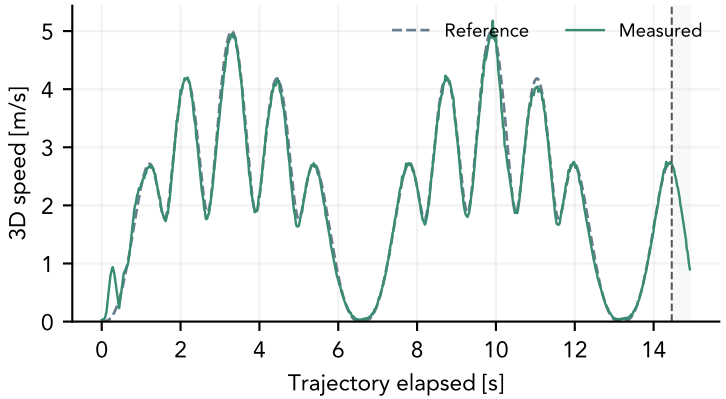
Raw error to selected reference



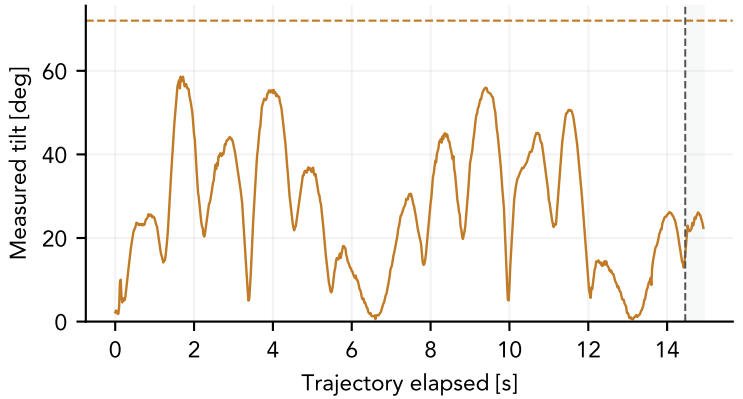
XYZ · solid measured / dashed reference



Recorded velocity state



Tilt · soft guard 72°



Per-eight RMS / max (cm): 1: 8.98 / 21.82 | 2: 7.85 / 12.75 | 3 partial: 4.21 / 5.27

MPC: 723 healthy ticks · 0 RTI rescues · 0 invalid | mocap age median / max 3.02 / 4.01 ms

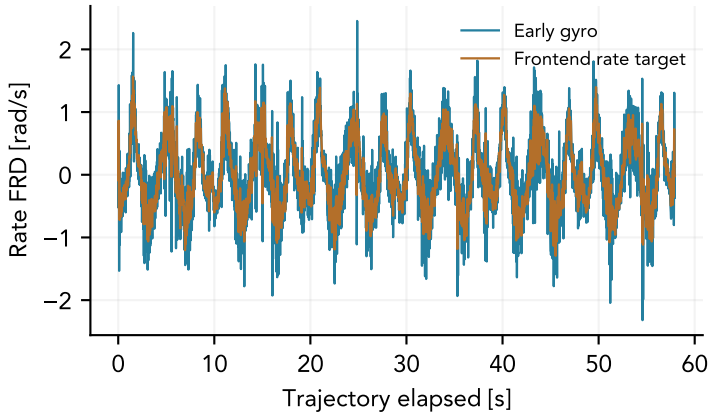
MPC tick-age abort at 14.460 s; 2 rad/s emergency brake. Hard world-height ceiling 1.5 m crossed 0.460 s later.

Actual recorded samples, no visualization filtering or time shift. State velocity uses the live causal estimator. FC native-rate audit follows separately.

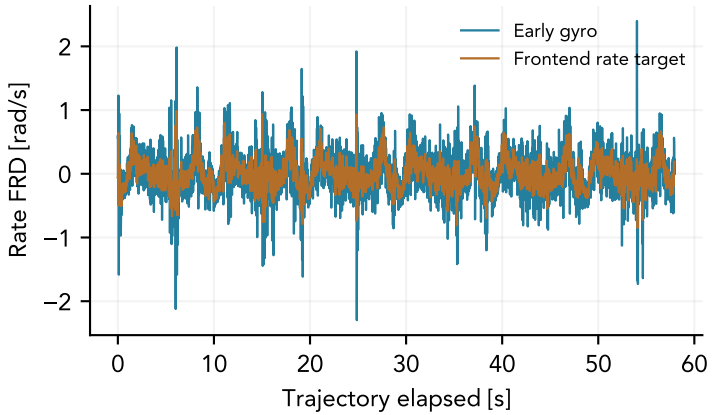
Appendix · MPC + PID 3 m/s · physical native measurements

Native mode PID throughout tracking; INDI weight zero. Rotor speed and applied counts are physical measurements, not unused INDI-shadow targets.

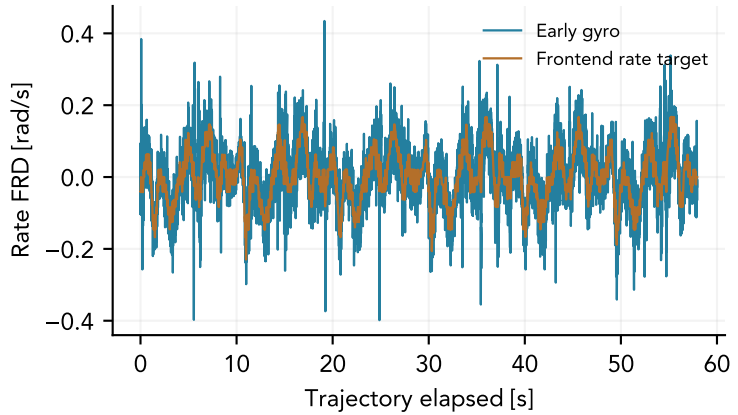
Roll · native physical rate



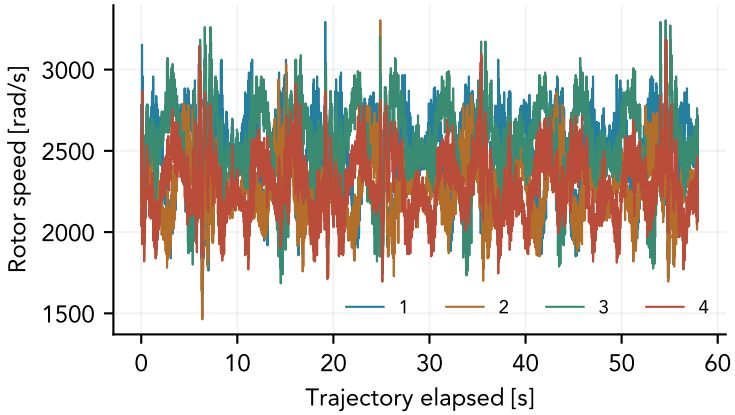
Pitch · native physical rate



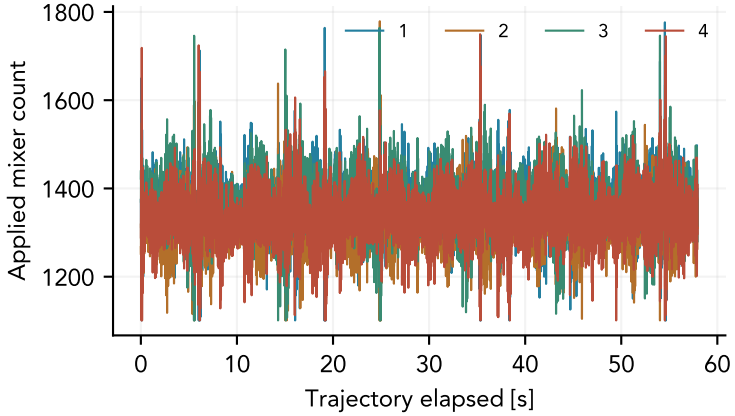
Yaw · native physical rate



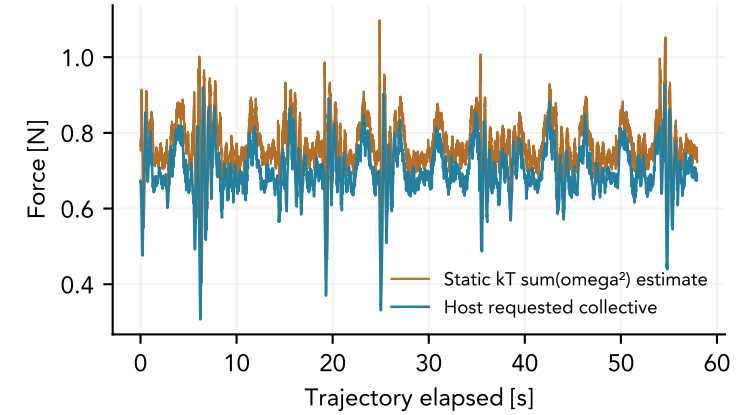
Actual measured rotor speed



Actual applied motor counts



Force request versus rotor-model estimate



Peak measured early gyro FRD [rad/s]: 2.453, 2.399, 0.435. Frontend target is getSetpointRate, not a rotor/servo shadow target.

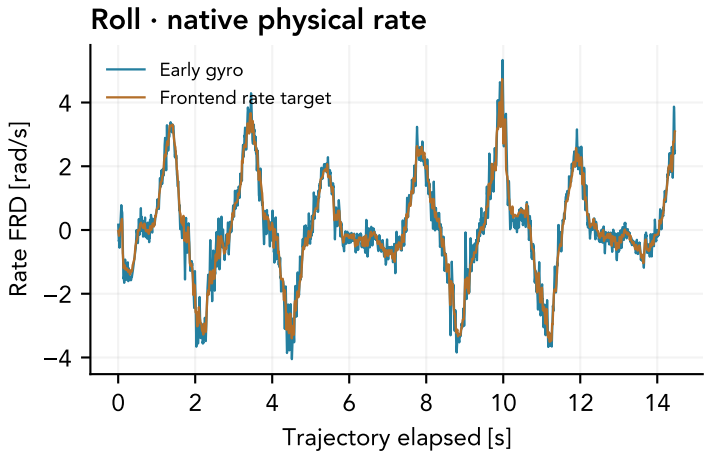
No PID actuator claims use INDI-shadow omega targets, servo errors, allocator residuals, desired alpha or torque. Internal PID feedback filtering differs from the early-gyro view.

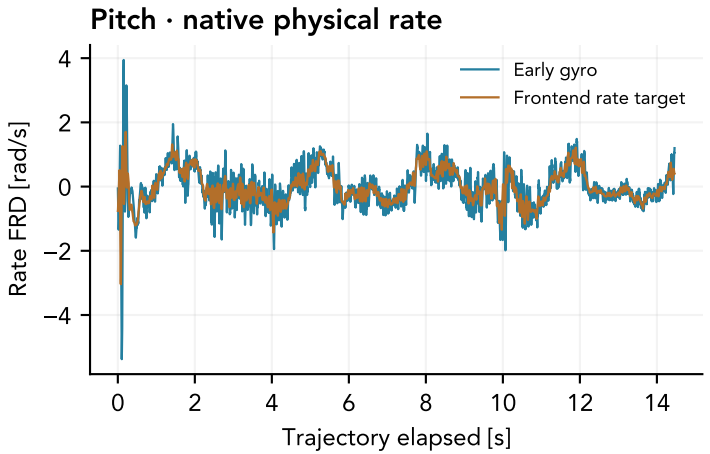
kT sum(omega²) is a calibrated static model estimate, not measured physical thrust. Dynamic inflow, voltage and mixer response can differ.

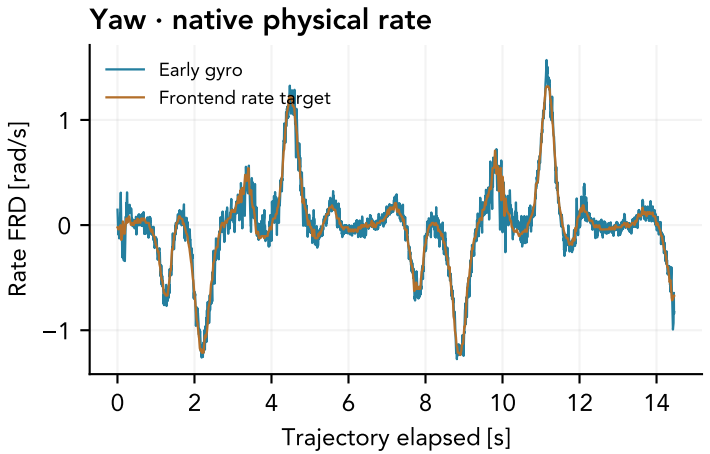
No added filter, resampling or fitted shift. Direct host/FC epoch mapping spread p5–p95: -57.32 to 58.03 ms.

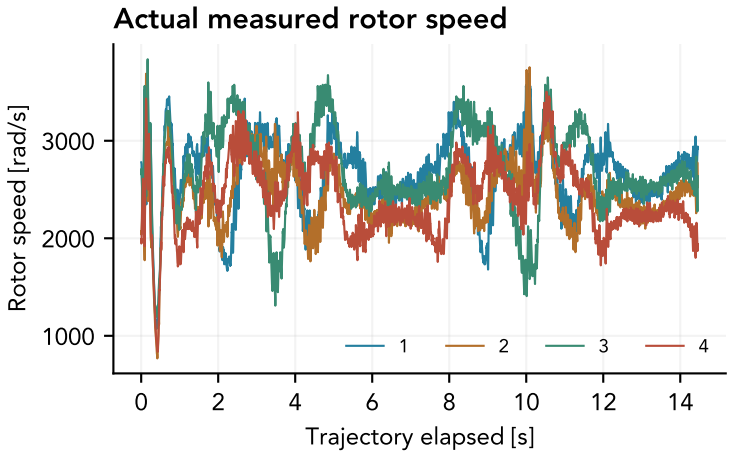
Appendix · MPC + PID 5 m/s · physical native measurements

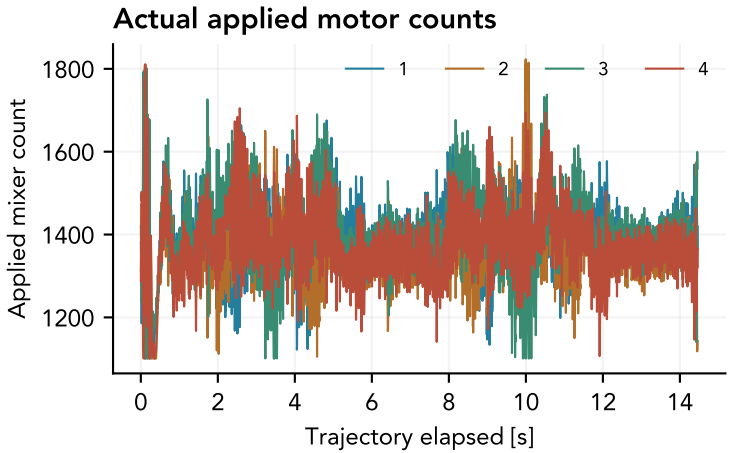
Native mode PID throughout tracking; INDI weight zero. Rotor speed and applied counts are physical measurements, not unused INDI-shadow targets.

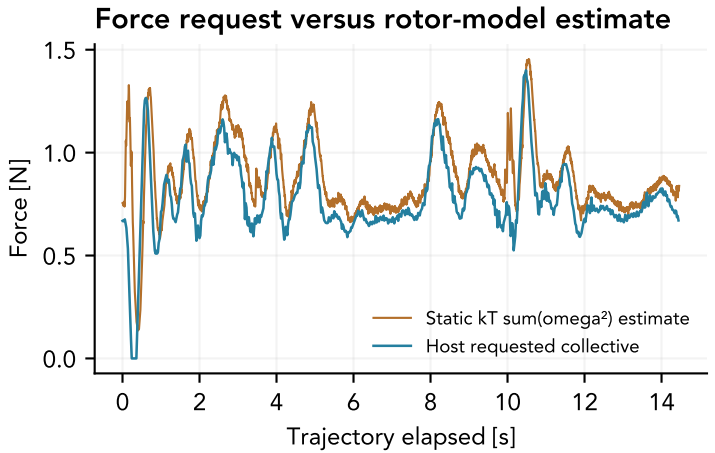












Peak measured early gyro FRD [rad/s]: 5.335, 5.376, 1.570. Frontend target is getSetpointRate, not a rotor/servo shadow target.

No PID actuator claims use INDI-shadow omega targets, servo errors, allocator residuals, desired alpha or torque. Internal PID feedback filtering differs from the early-gyro view.

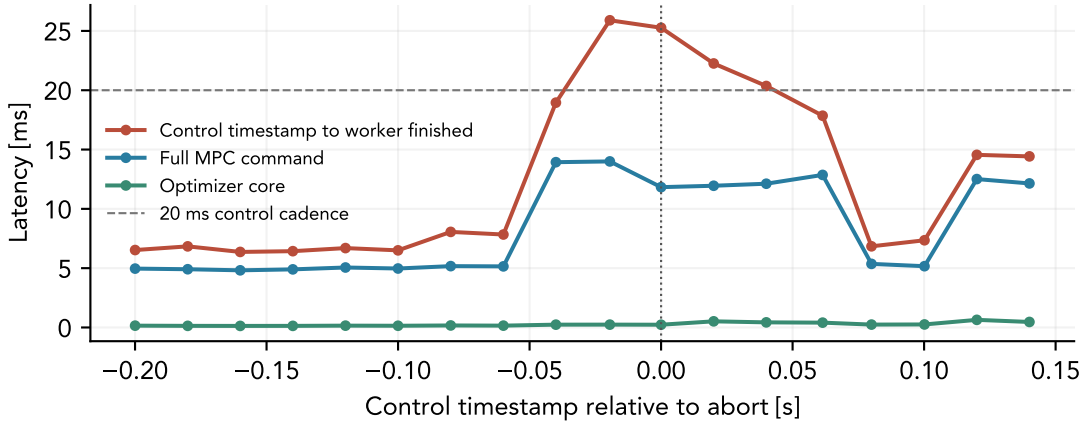
kT sum(omega²) is a calibrated static model estimate, not measured physical thrust. Dynamic inflow, voltage and mixer response can differ.

No added filter, resampling or fitted shift. Direct host/FC epoch mapping spread p5–p95: -21.13 to 19.84 ms.

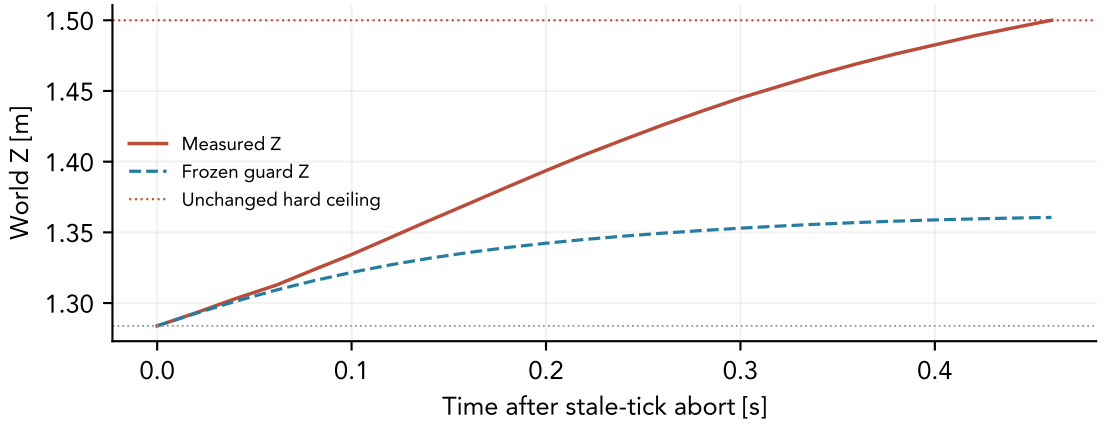
PID 5 m/s · stale tick triggered braking; rising height triggered motor stop

Native PID stayed fault-free. The actual 2 rad/s EmergencyLanding was active; it had not reached a settled anchor when the 1.5 m ceiling tripped.

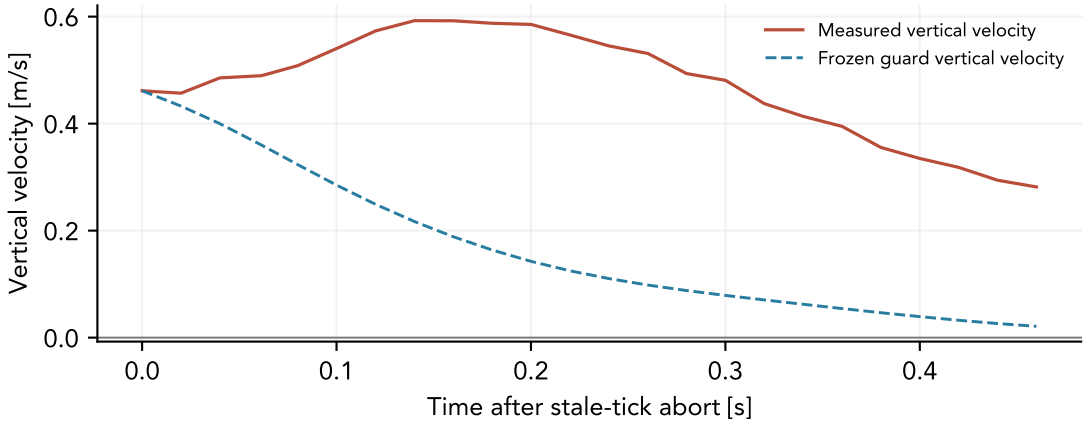
Freshness loss · valid result arrived late



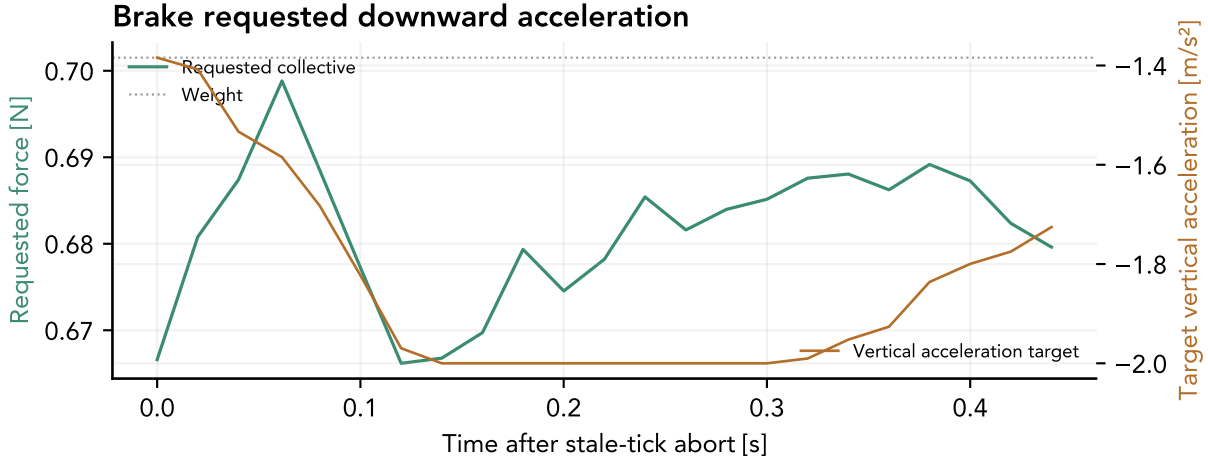
Hard ceiling crossed during emergency brake



Actual upward momentum exceeded prediction



Brake requested downward acceleration



Tick: core / solve block / full command / control-time to finished [ms] 1676: 0.24 / 1.13 / 13.93 / 18.95 | 1677: 0.24 / 1.18 / 14.00 / 25.90 | 1678: 0.23 / 1.13 / 11.83 / 25.27

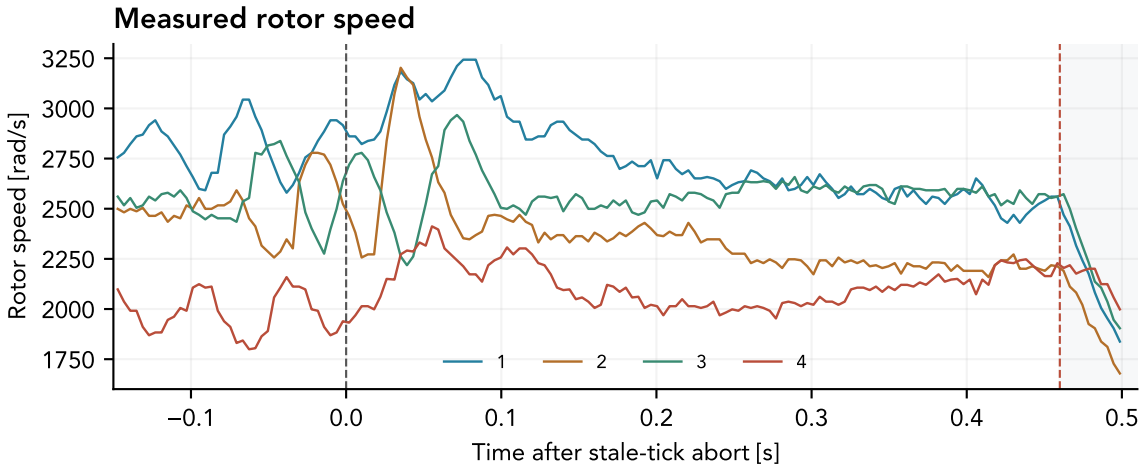
Initial Z 1.2838 m, upward velocity 0.461 m/s; Z reached 1.500002 m after 0.460 s. The vertical response mismatch is separate from the stale-result trigger.

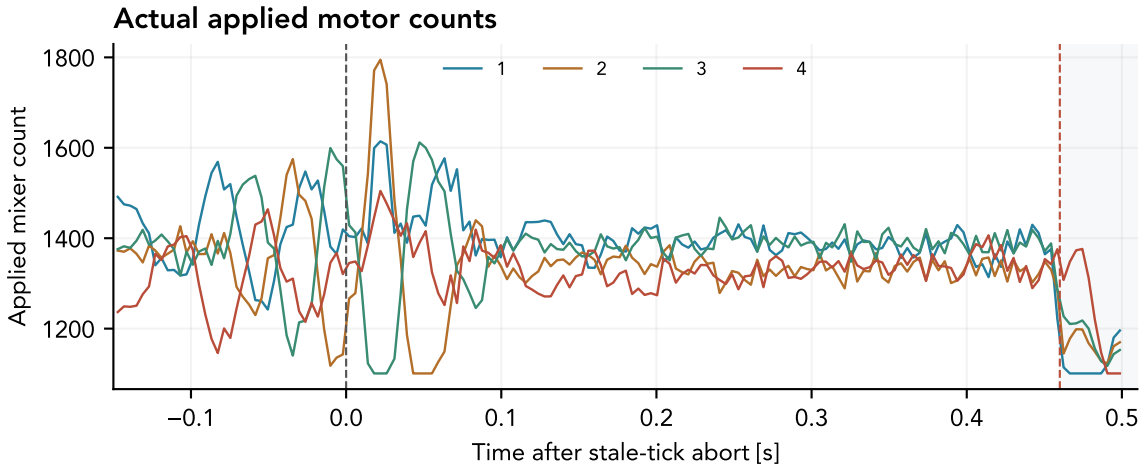
Control-time latency excludes pose age. Most time was outside the optimizer core; queue, GIL, IPC and scheduling are not individually profiled.

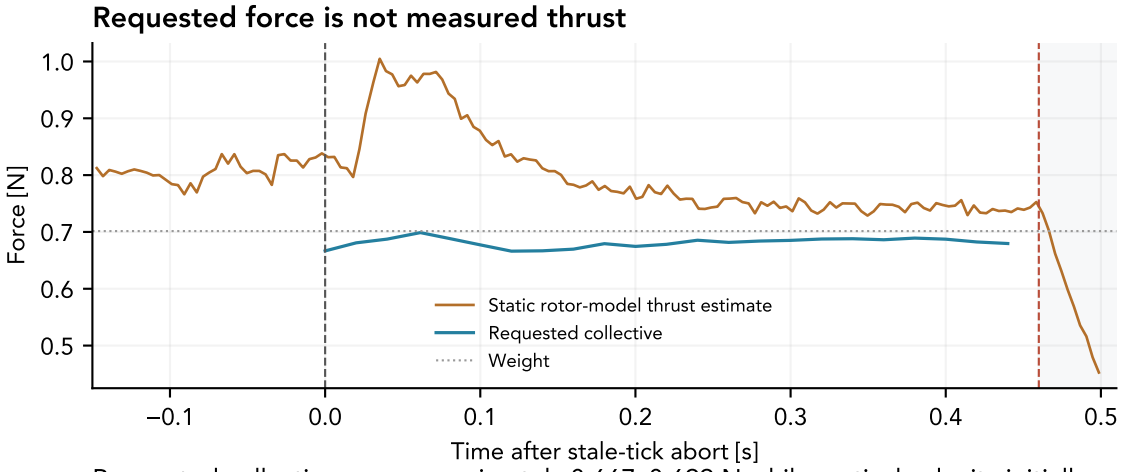
No added filters or time shifts. Later solver infeasibility occurred after the emergency reference changed and did not initiate this abort.

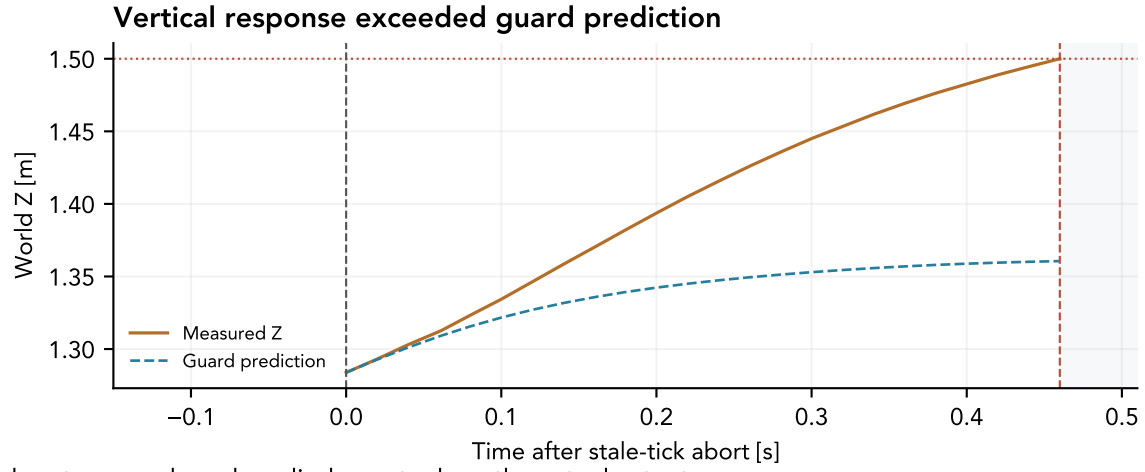
Appendix · MPC + PID 5 m/s · actual motor response during emergency braking

Dashed lines: stale-tick emergency entry and host hard-altitude stop. Native PID remains selected; there is no INDI allocator takeover or fault.









Requested collective was approximately 0.667–0.699 N while vertical velocity initially rose. Measured motor speeds and applied counts show the actual actuator response.

The rotor-based force estimate is static and model-derived; it supports checking collective-response mismatch but does not identify mixer / airmode as the cause.

Next step: characterize the PID vertical / collective transient and braking ceiling margin; profile the full 20 ms result pipeline separately from the optimizer.

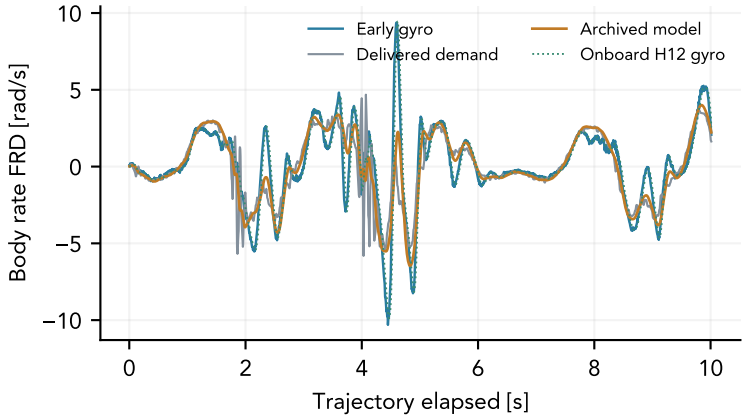
No added filters, resampling or fitted shifts. Native clock alignment has finite packet-age uncertainty; shaded samples follow the host stop observation.

5 m/s butterfly · native-rate audit · allocator fault

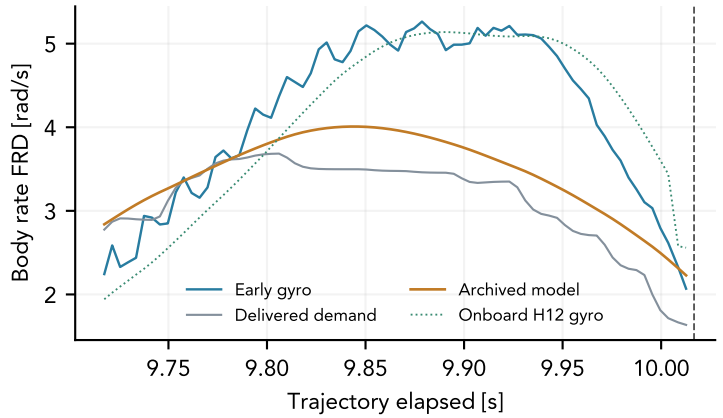
Archive matched by FC time / sequence / mode. Recorded early gyro and delivered demand in FRD; onboard H12 shown separately. No added filters or shifts.

Roll gyro / model peak 10.310 / 6.476 rad/s | archived roll model wn 19.540 rad/s, damping 0.209 | no extra transport delay

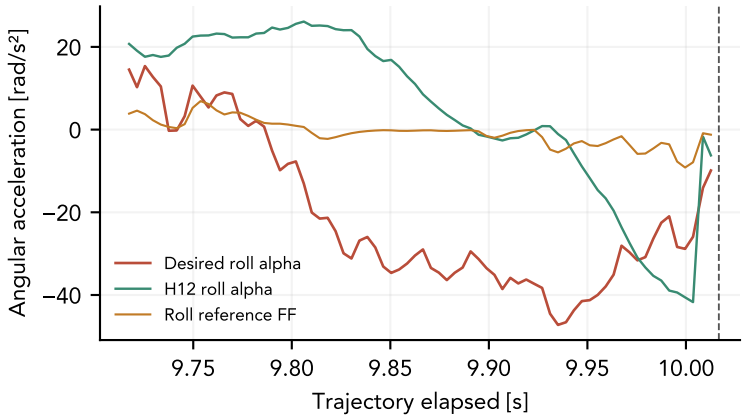
Roll · full healthy INDI tracking



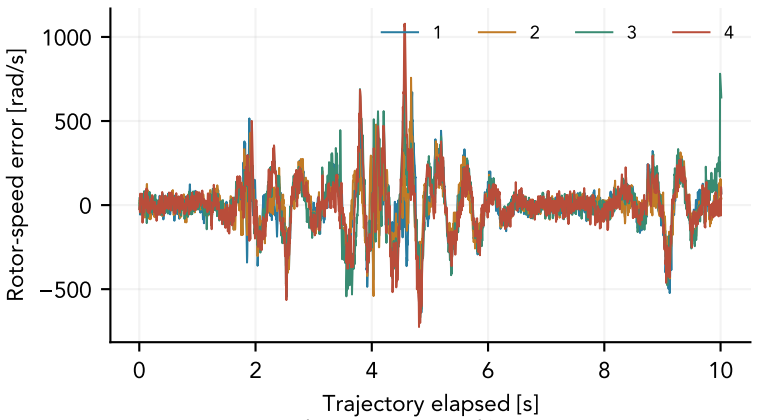
Roll · native fault approach



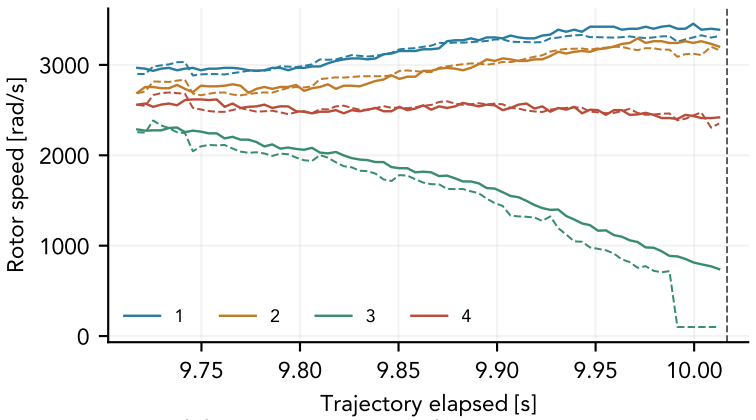
INDI angular-acceleration terms



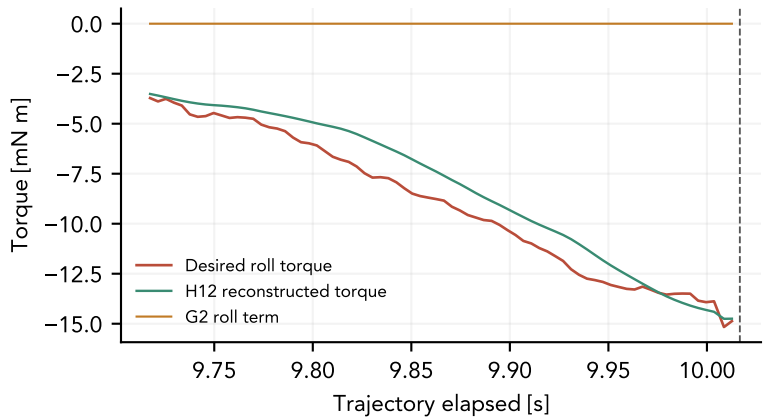
Native RPM feedback · raw minus target



Rotor speed · solid raw / dashed target



INDI roll torque · before fallback only



Raw motor-speed RMS error [rad/s], motors 1–4: 140.7, 127.5, 162.8, 164.2. Model gyro RMS error [rad/s]: 1.202, 0.445, 0.100.

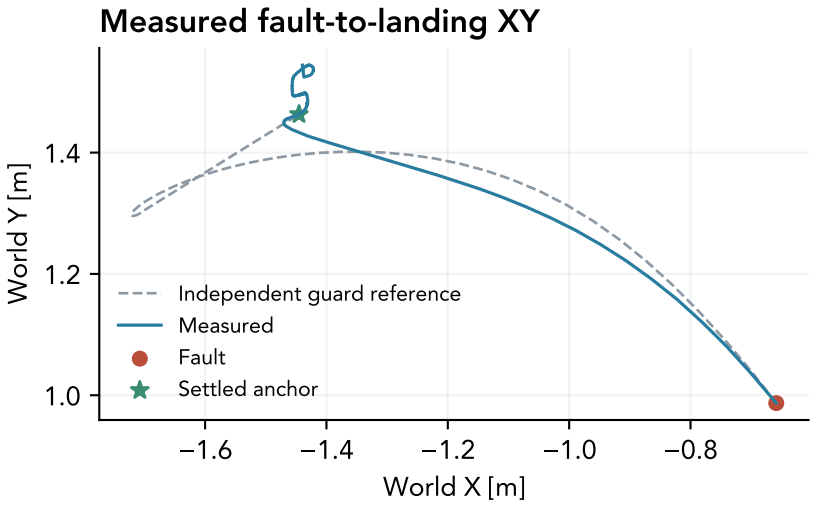
First sampled fallback: FC 81160944 us; prior solver 1 (bounded), saturation mask 4. The initiating 1 kHz rejection was not retained at 250 Hz.

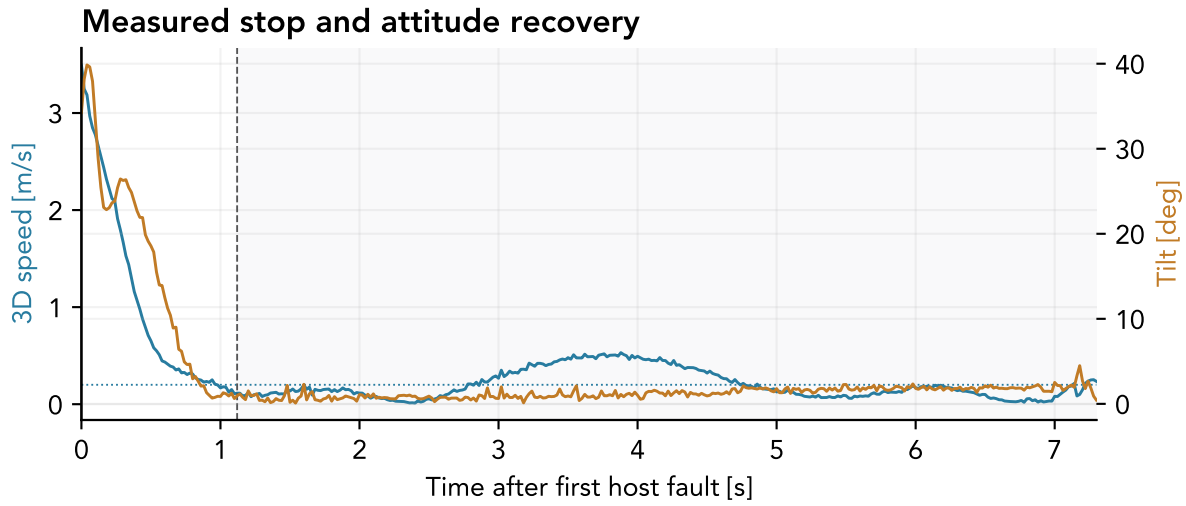
Native deadline / late increments: +1 each before fallback; +3 each through host abort. Timing causality remains unresolved.

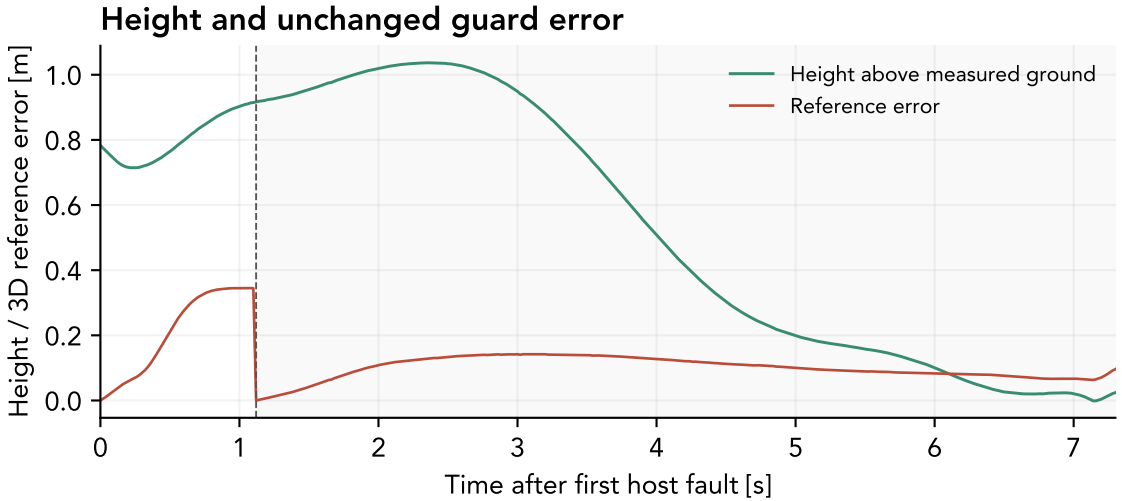
Motor torque is model-reconstructed. Allocator / desired-alpha / torque shadow fields are not interpreted as PID actuation after native fallback.

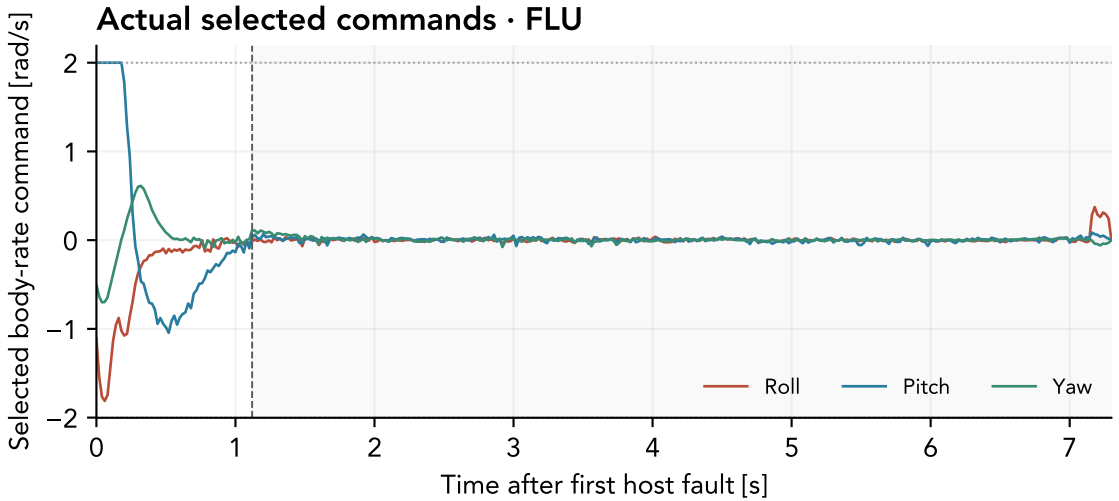
5 m/s fault recovery · emergency brake completed, trajectory remained partial

Native allocator fault initiated fallback; host MPC remained healthy. Actual emergency cap 2 rad/s, ordinary takeoff / settled landing 0.6 rad/s.









Speed below 0.2 m/s after 0.98 s; settled anchor latched after 1.12 s. Maximum XY displacement 0.960 m; reference error 0.345 m.

Braking height above ground 0.714–0.917 m; landing ended 7.30 s after fault. Shutdown acknowledged; no recorder loss.

All recorded samples retained; no added filtering, interpolation or time shift. Shading marks descent after settled-anchor latch.